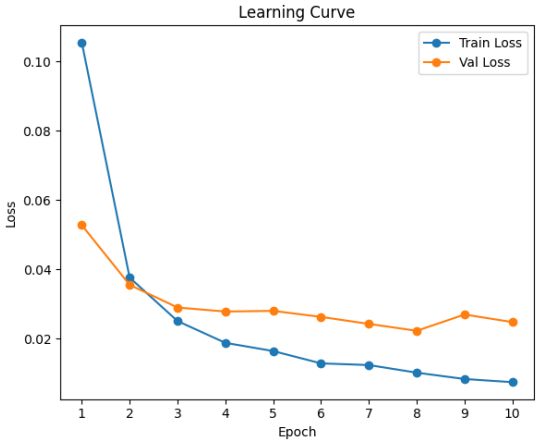
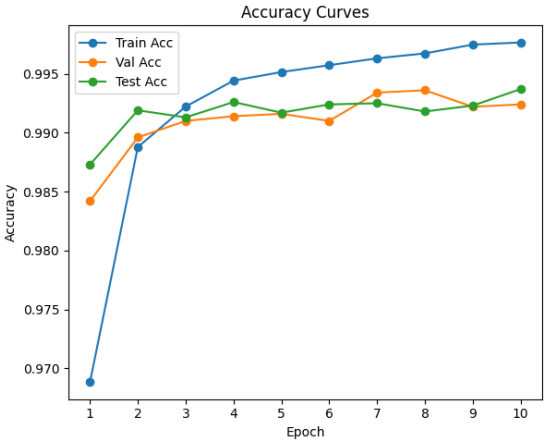
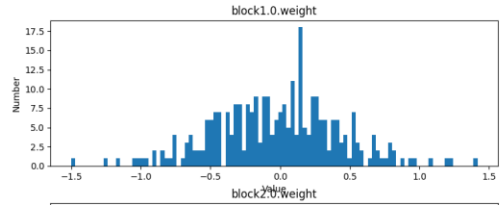
I. MNIST


Learning Curve
Train Loss
Val Loss
Loss
Epoch


Accuracy Curves
Train Acc
Val Acc
Test Acc
Accuracy
Epoch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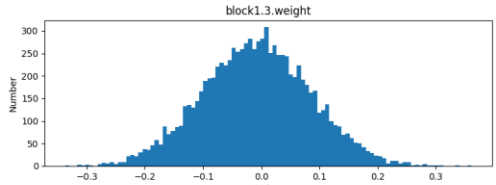
Number
Value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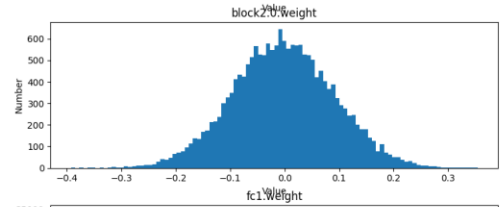
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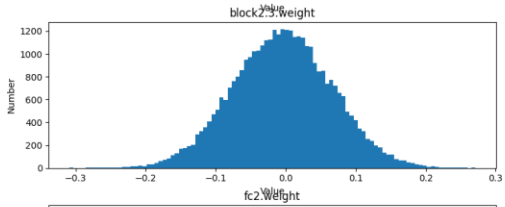
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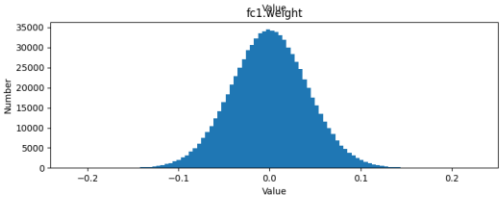
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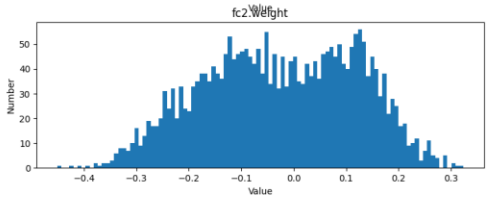
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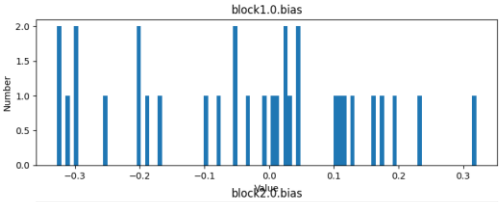
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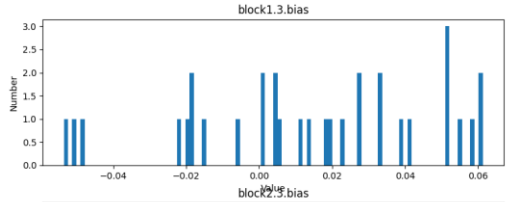
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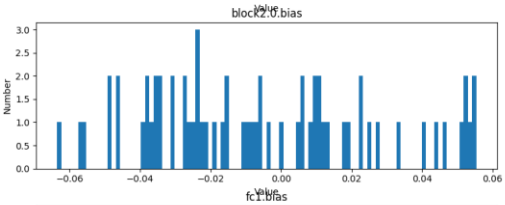
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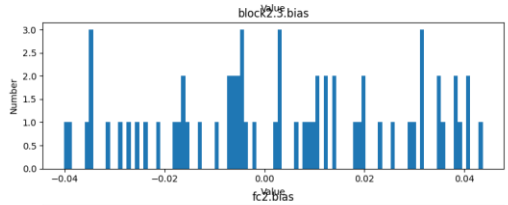
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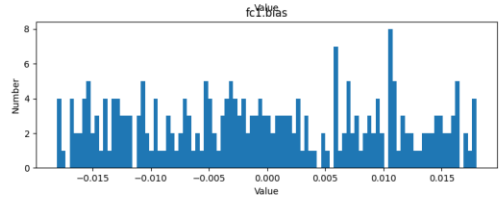
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Number
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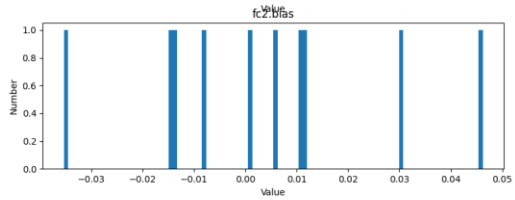
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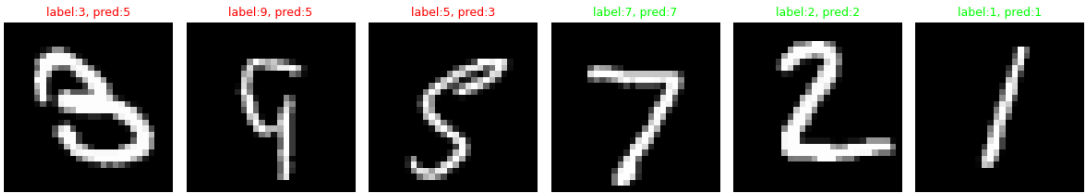
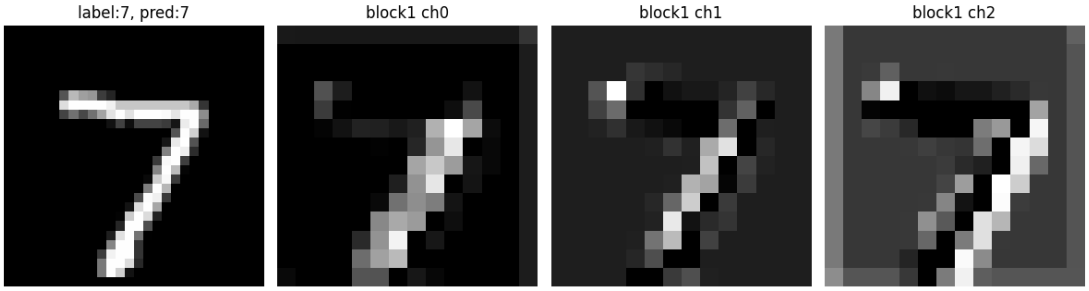
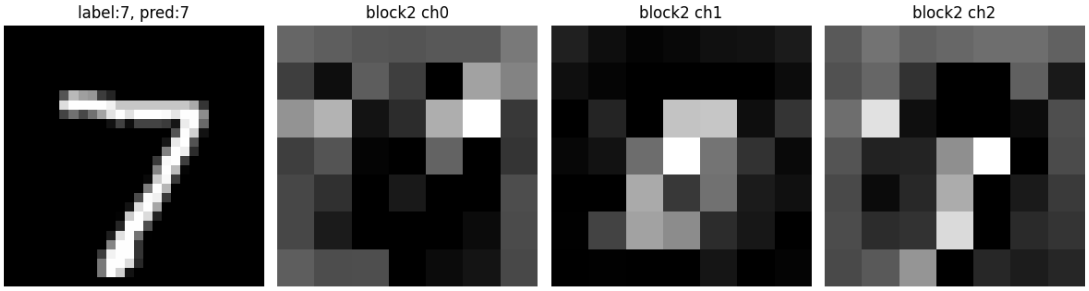
Number
Value







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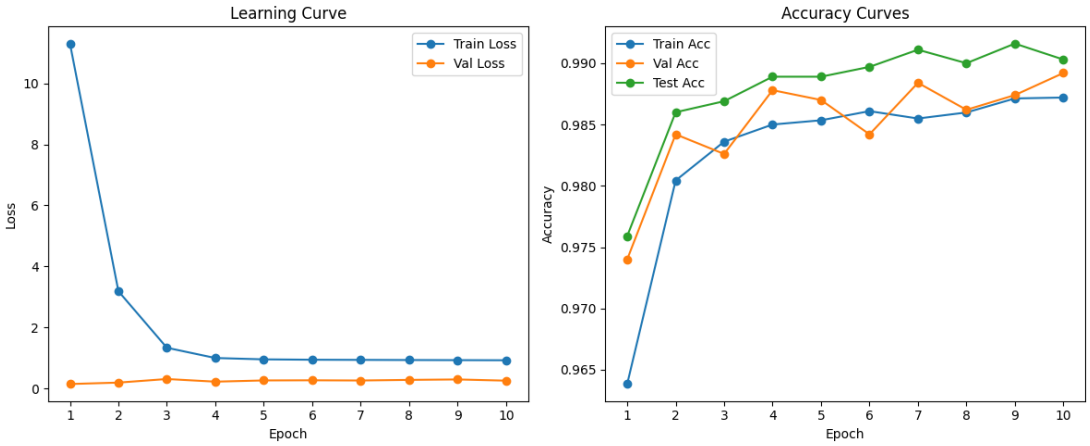

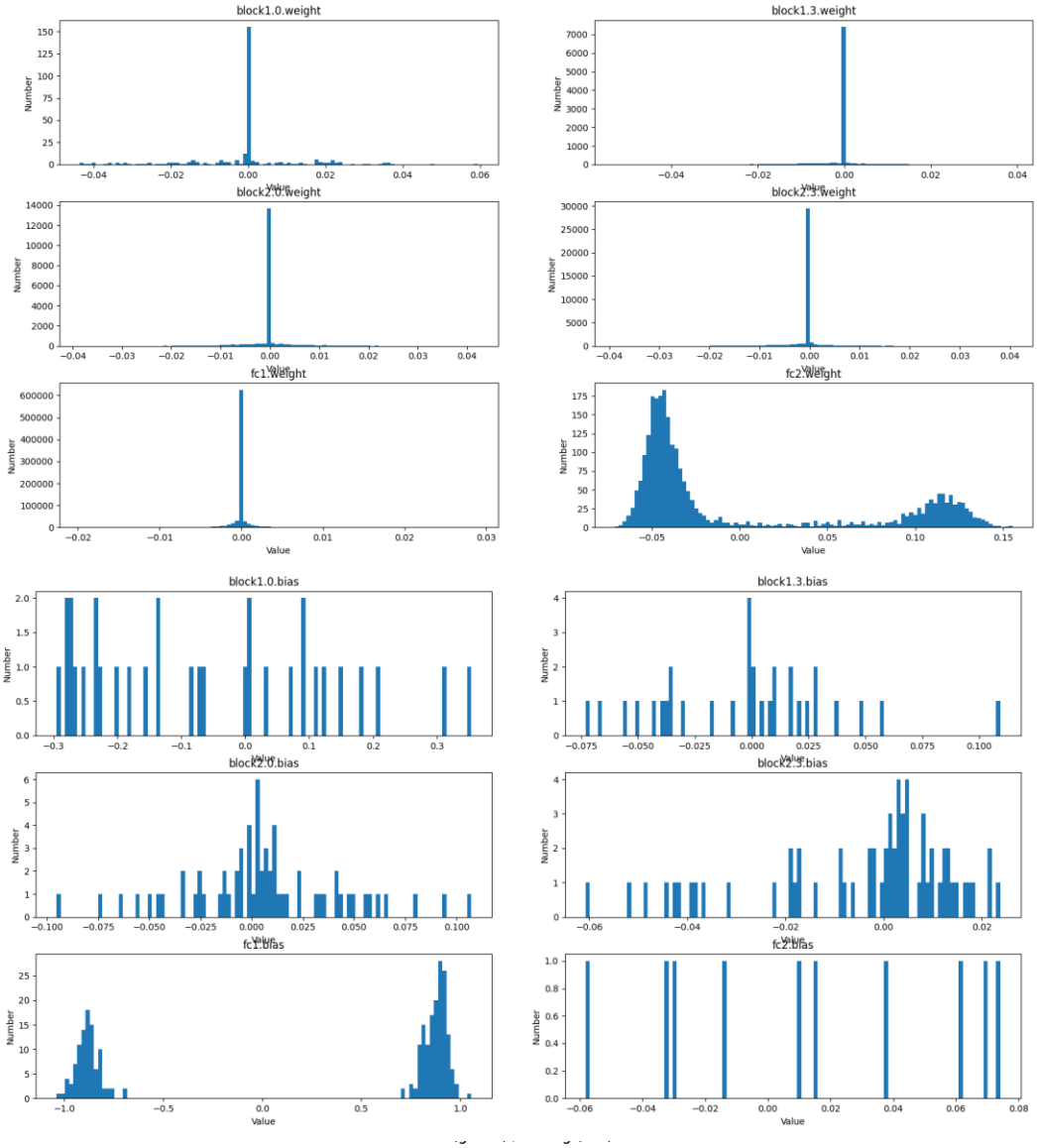
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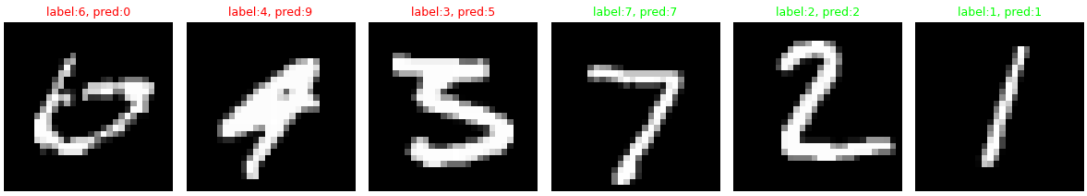
block1.3.weight
block2.0.weight
block2.3.weight
fc1.weight
fc2.weight
block1.0.bias
block1.3.bias
block2.0.bias
block2.3.bias
fc1.bias
fc2.bias
Number
Value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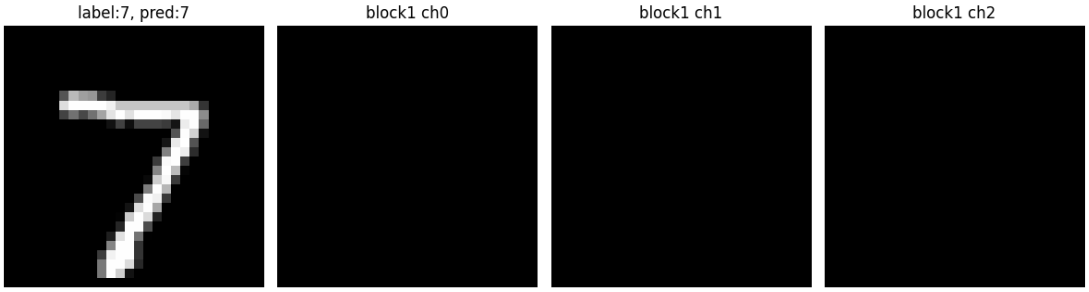
label:4, pred:9
label:3, pred:5
label:7, pred:7
label:2, pred:2
label:1, pred:1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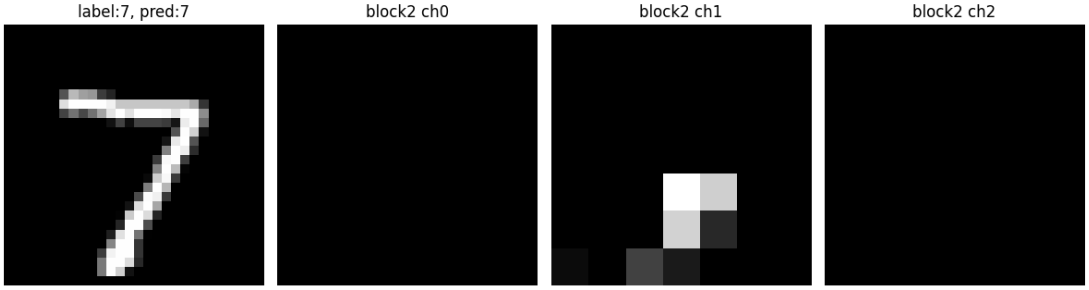
block1 ch0
block1 ch1
block1 ch2



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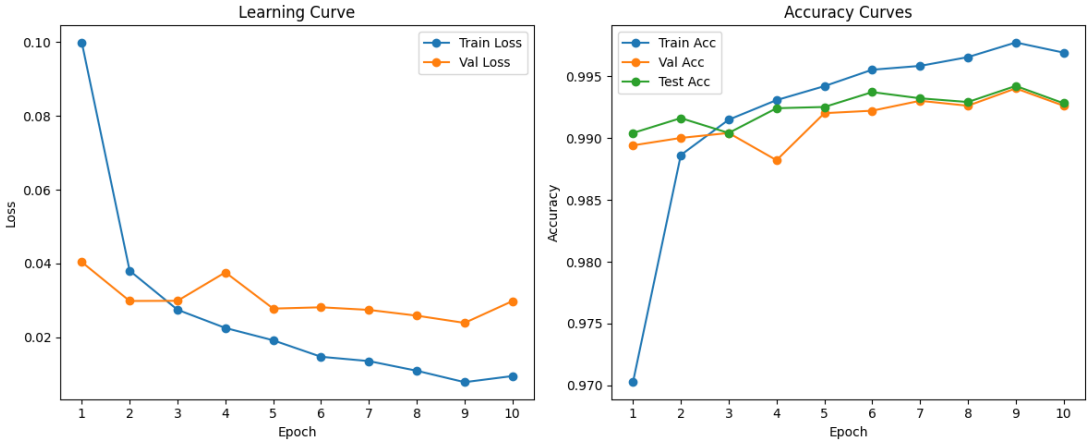

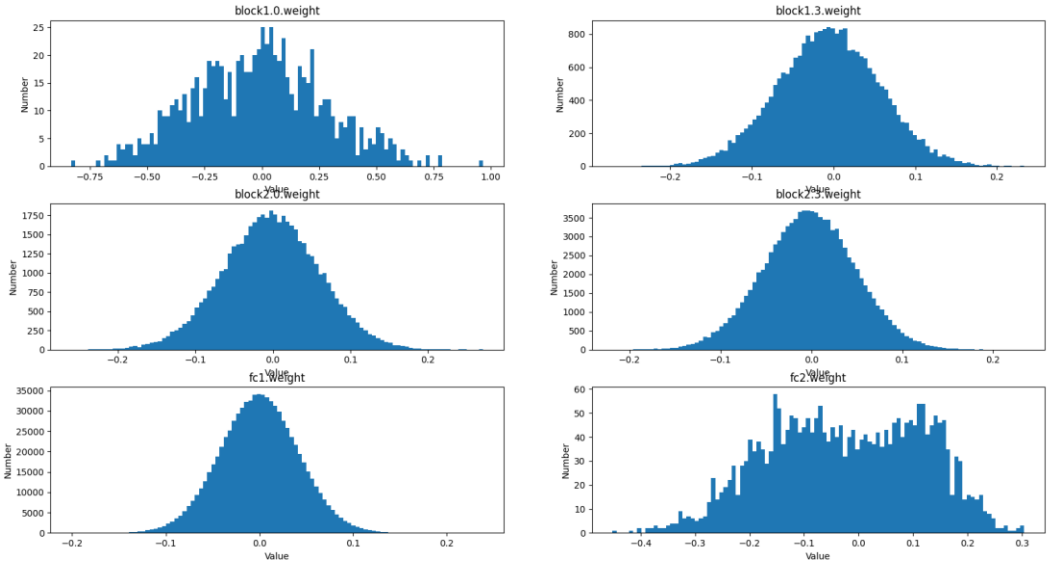

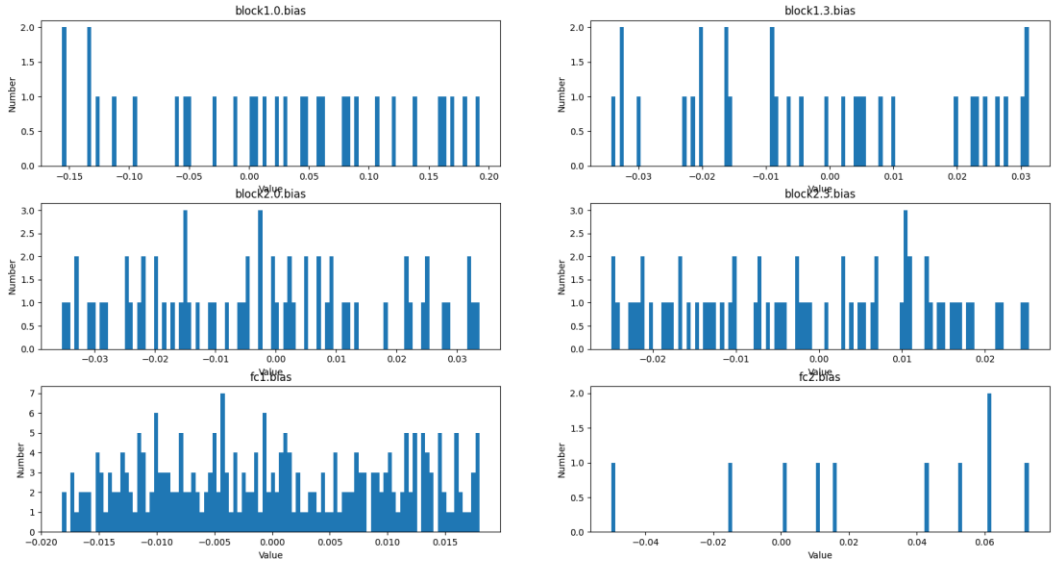

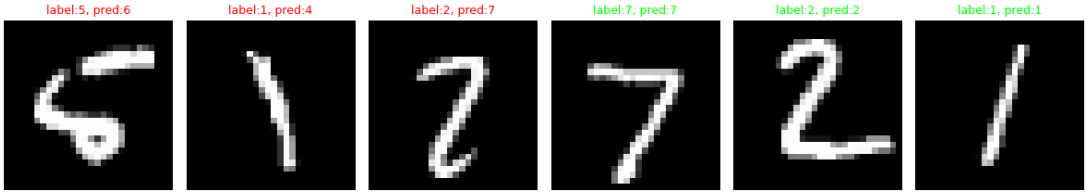

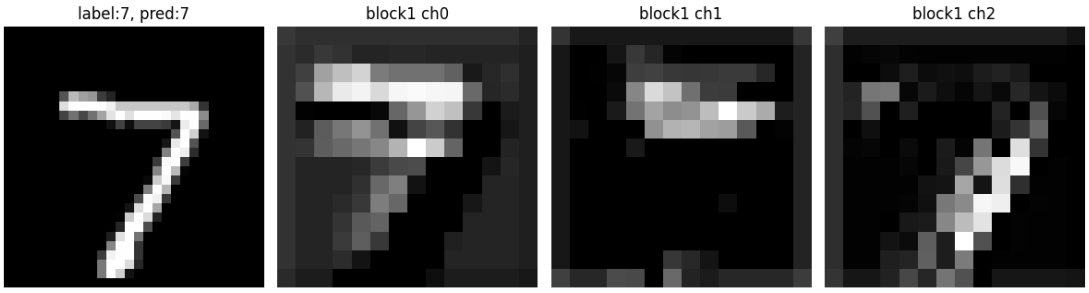

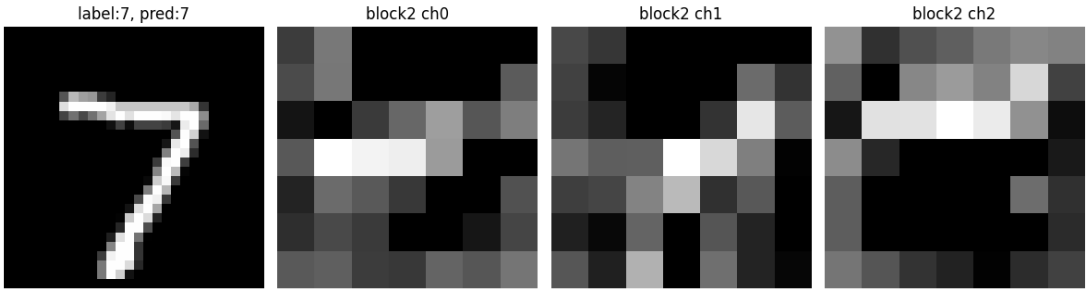

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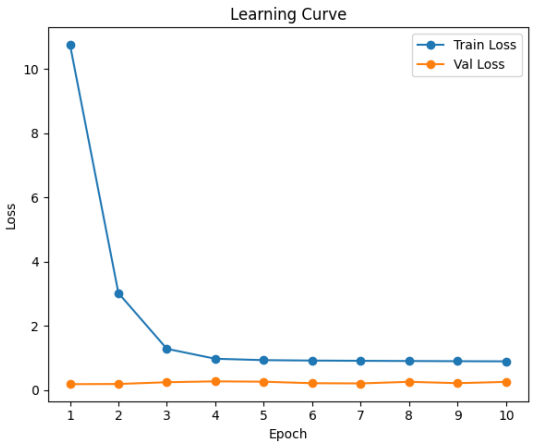
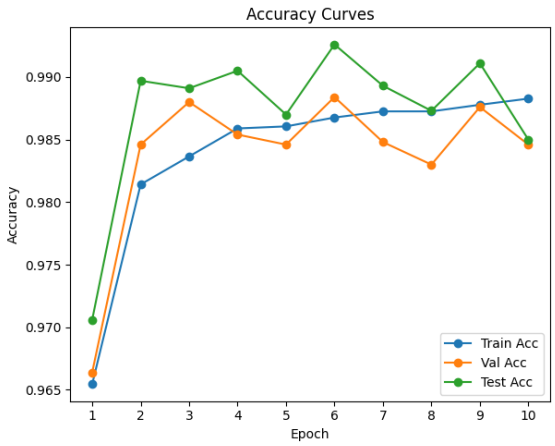
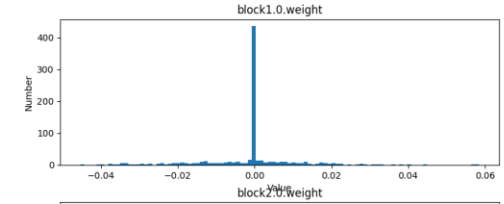

Learning Curve
Train Loss
Val Loss
Loss
Epoch


Accuracy Curves
Train Acc
Val Acc
Test Acc
Accuracy
Epoch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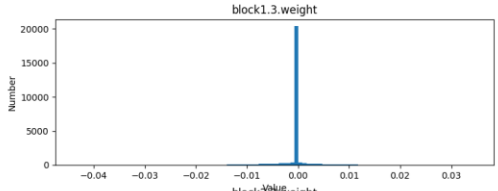
Value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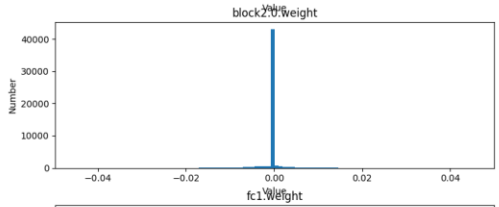
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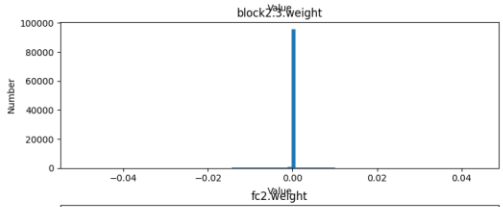
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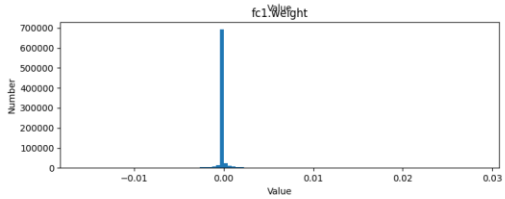
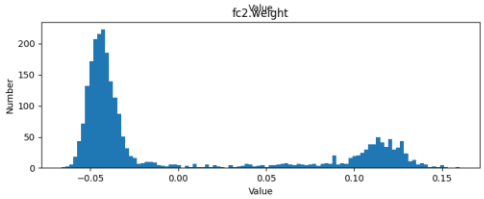

fc1.weight
Number
Value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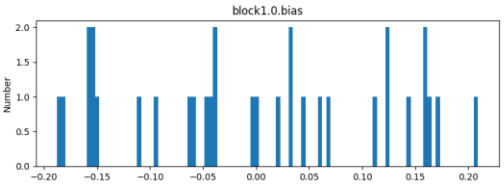
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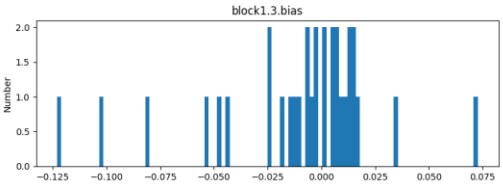
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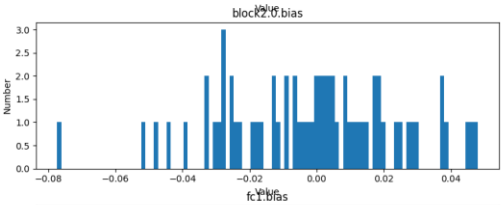
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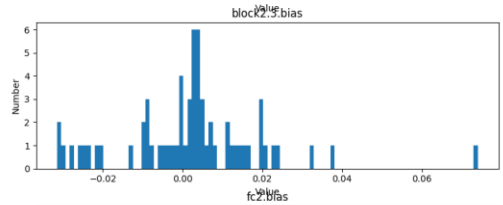
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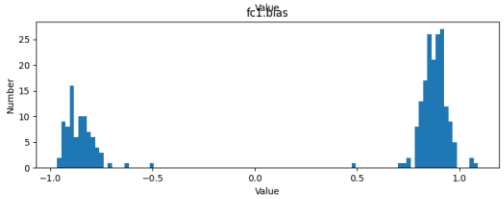
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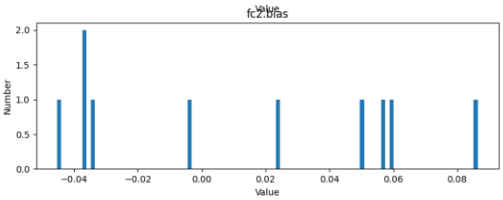
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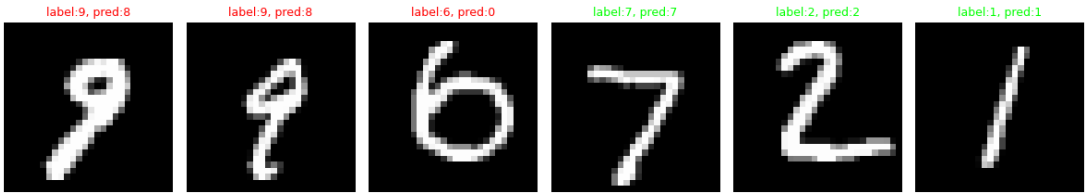
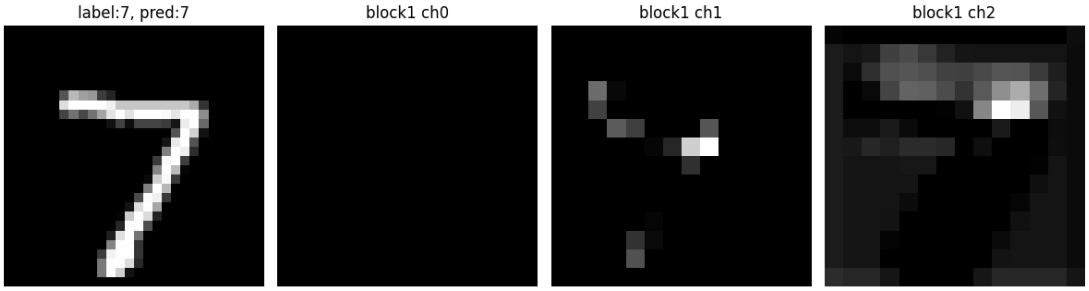
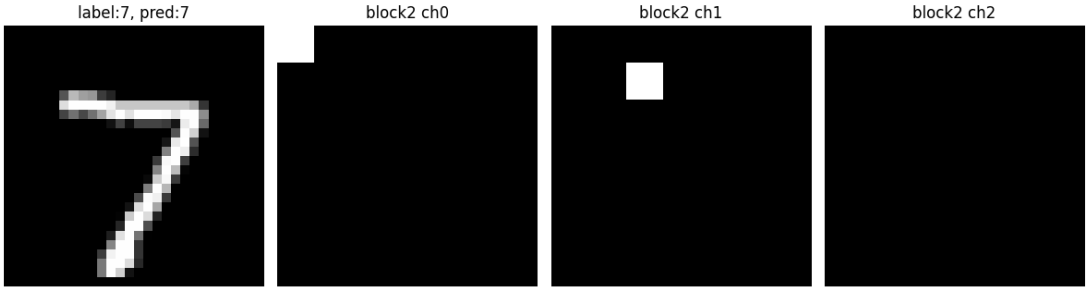
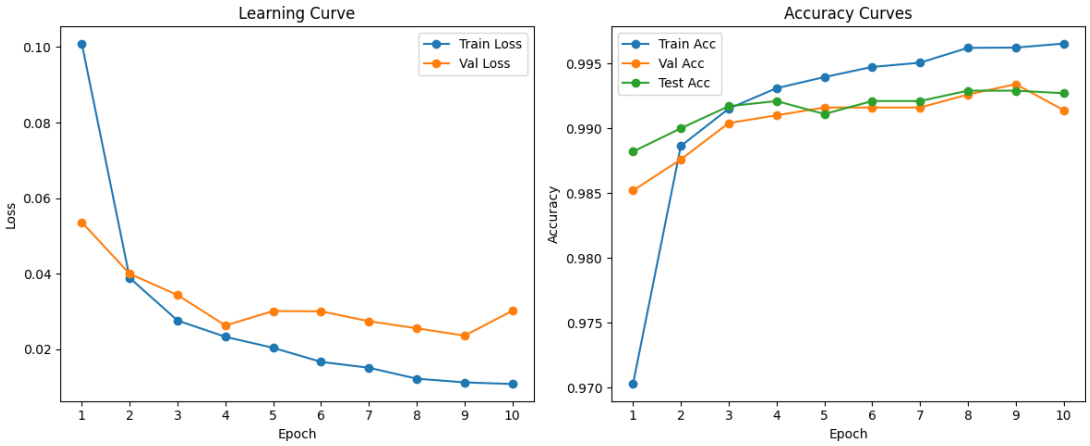
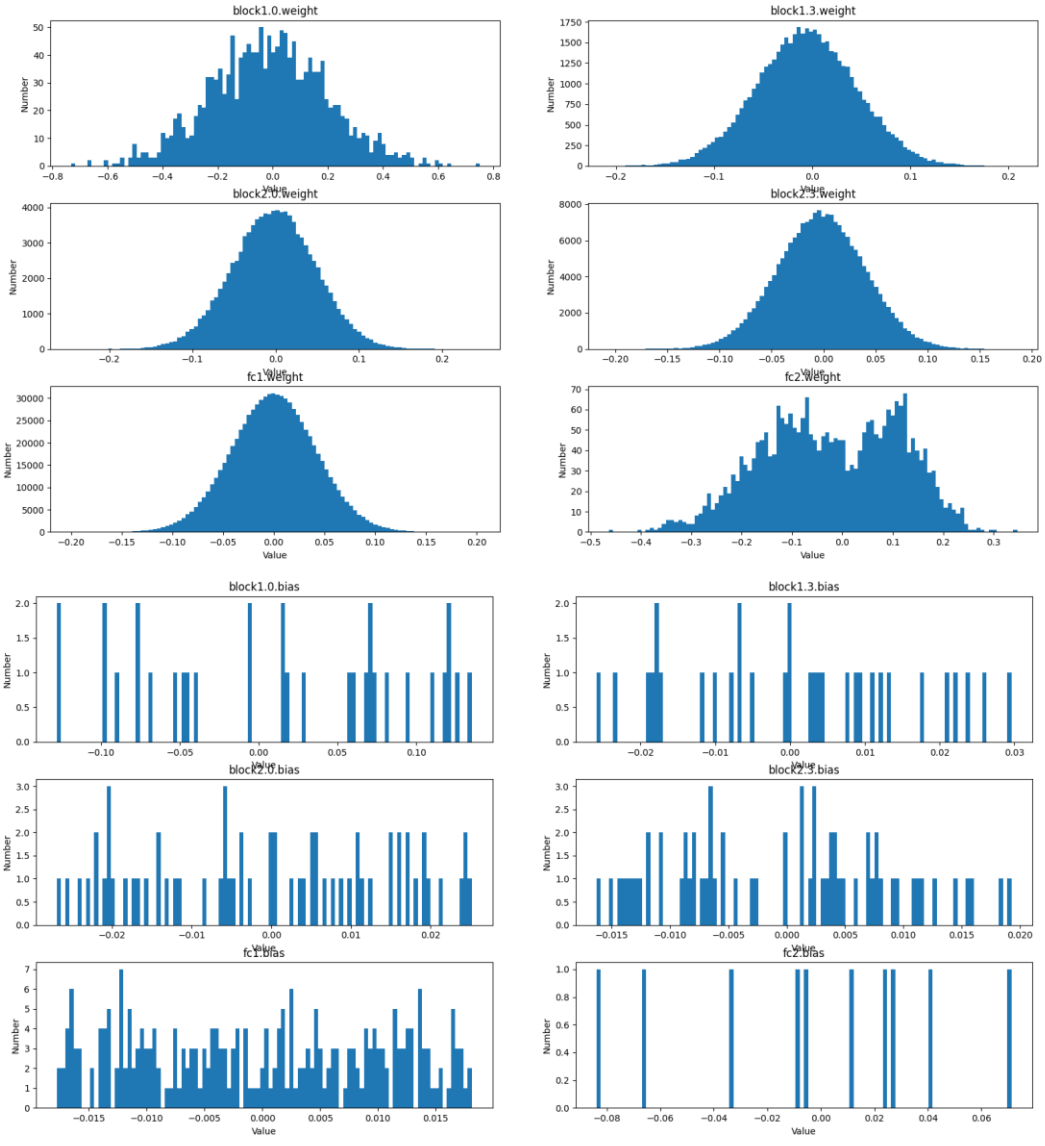


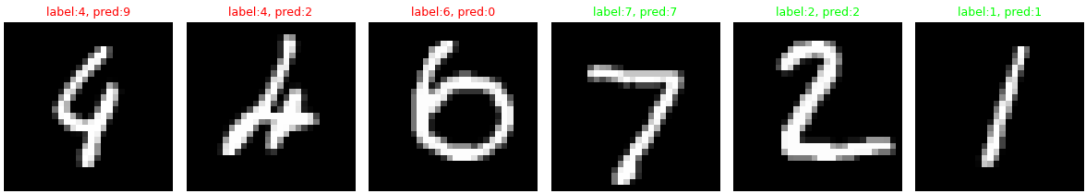




Kernel = 5, *Padding* = 2, *Stride* = 1, *L2 rate* = 5e-2




block1.0.weight
block1.3.weight
block2.0.weight
block2.3.weight
fc1.weight
fc2.weight
block1.0.bias
block1.3.bias
block2.0.bias
block2.3.bias
fc1.bias
fc2.bias
Number
Value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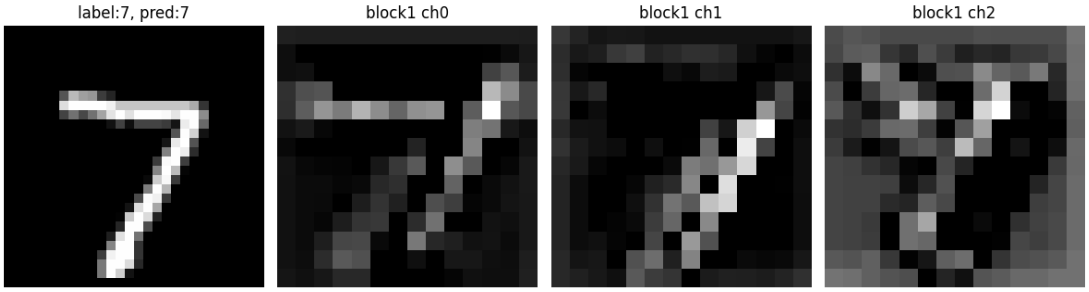
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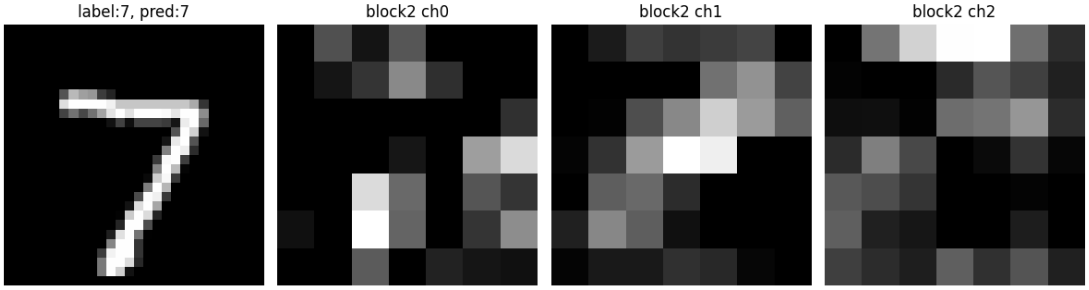
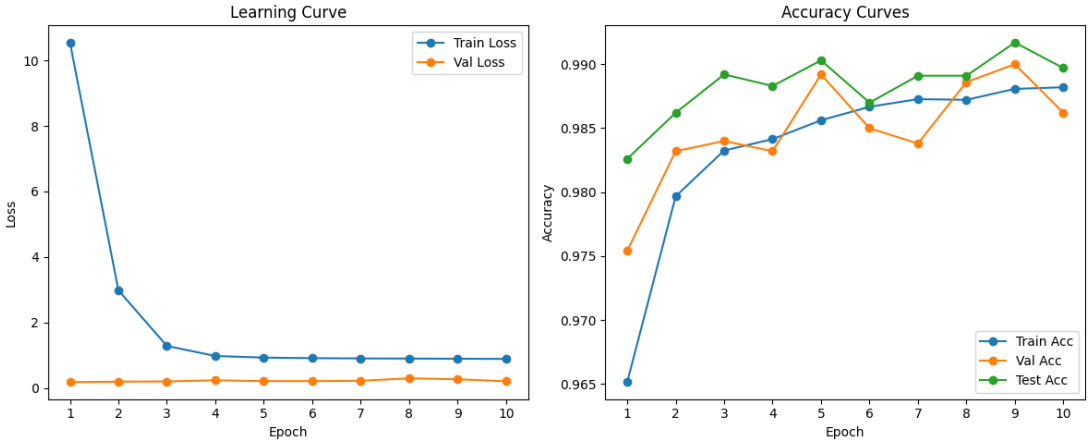
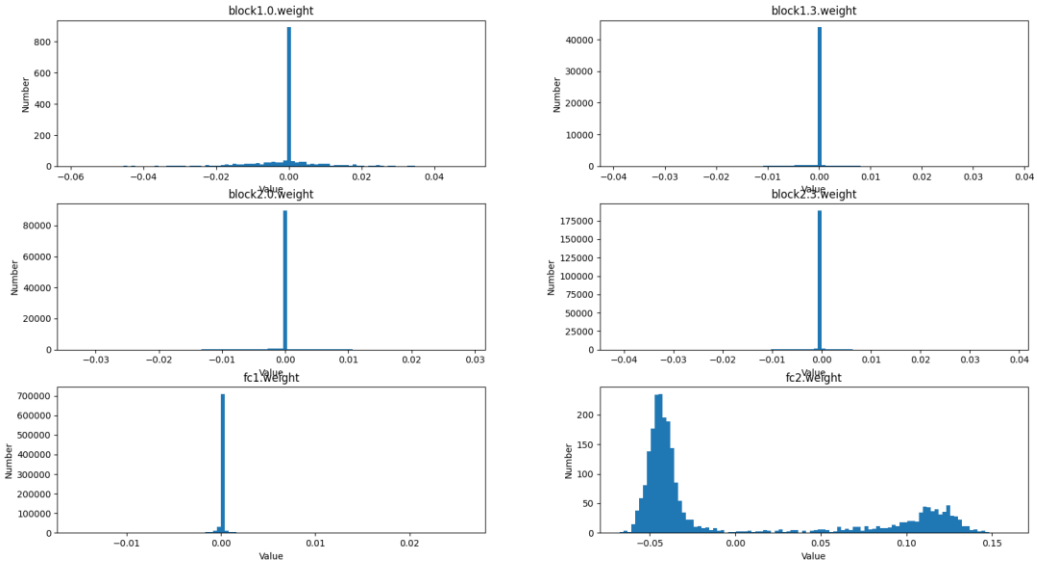
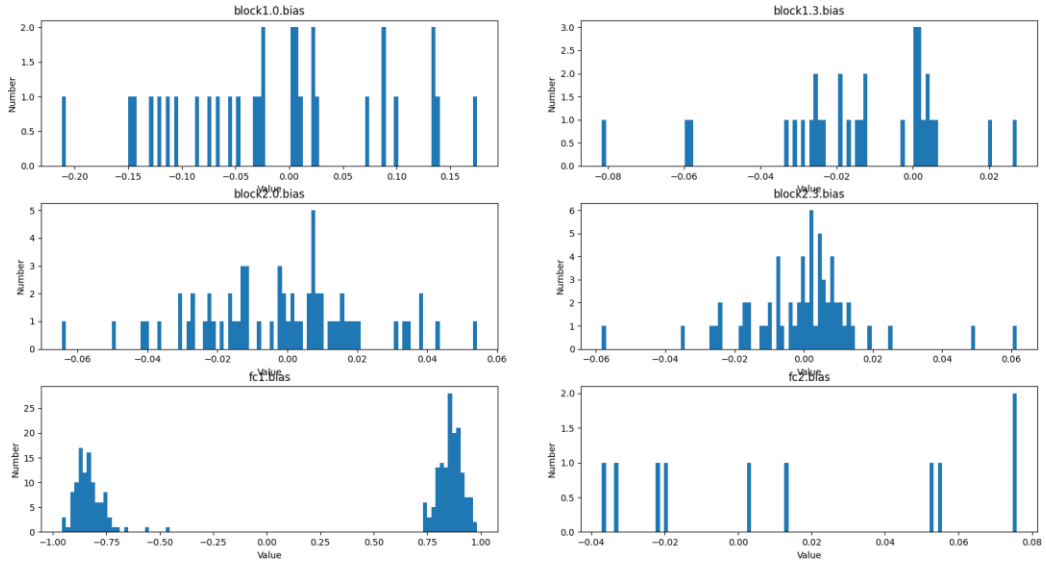
block1 ch0
block1 ch1
block1 ch2

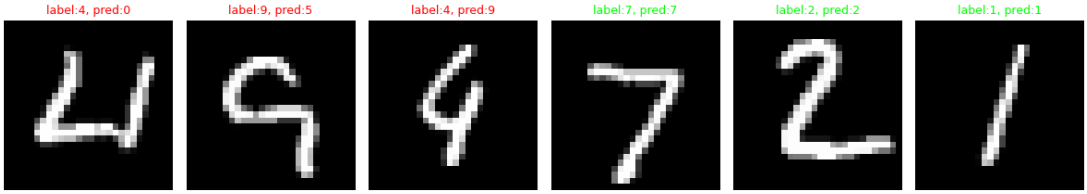


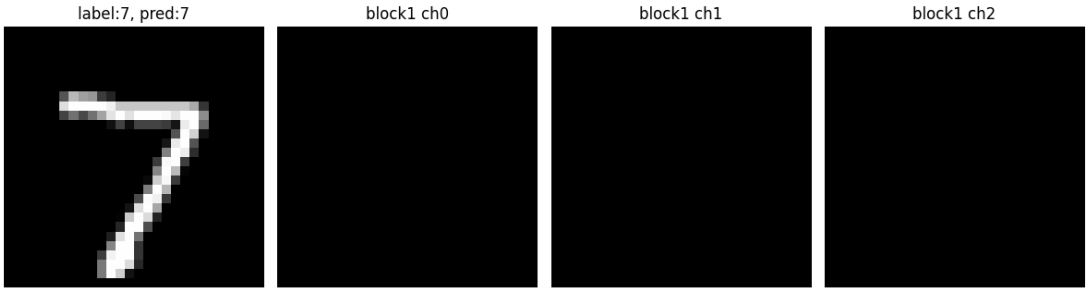
Kernel = 7, *Padding* = 3, *Stride* = 1, *L2 rate* = 0

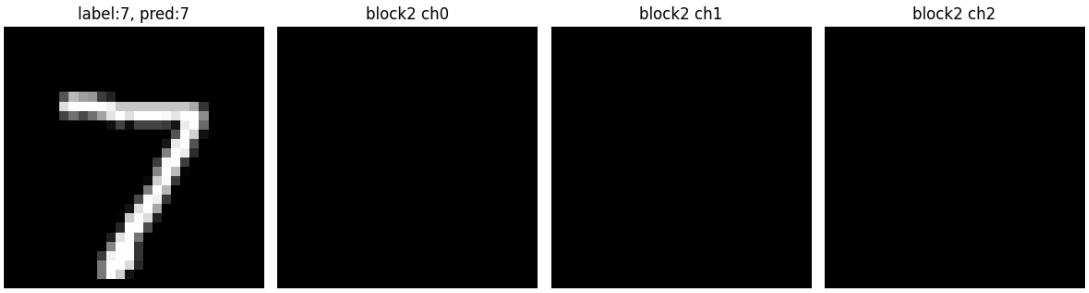












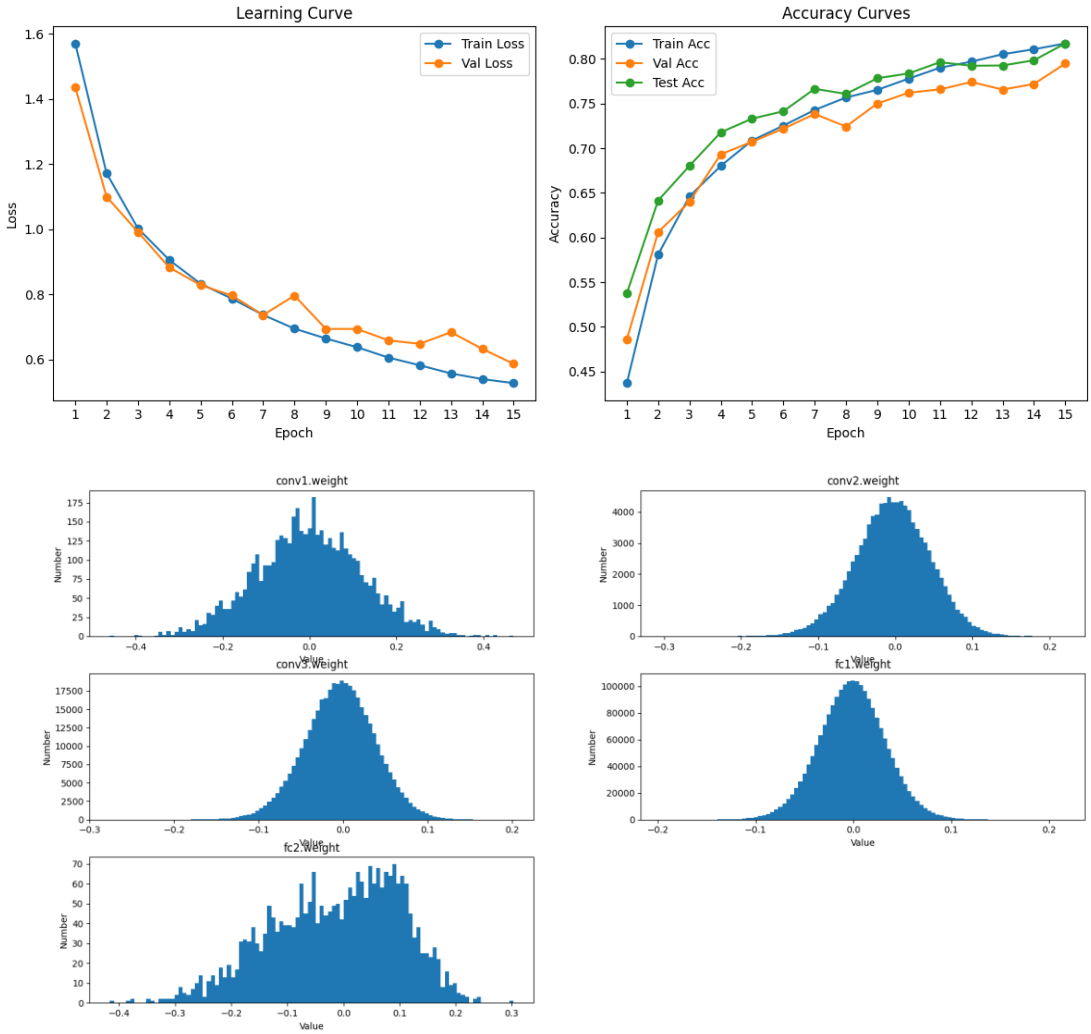
Kernel = 7, *Padding* = 3, *Stride* = 1, *L2 rate* = 5e-2

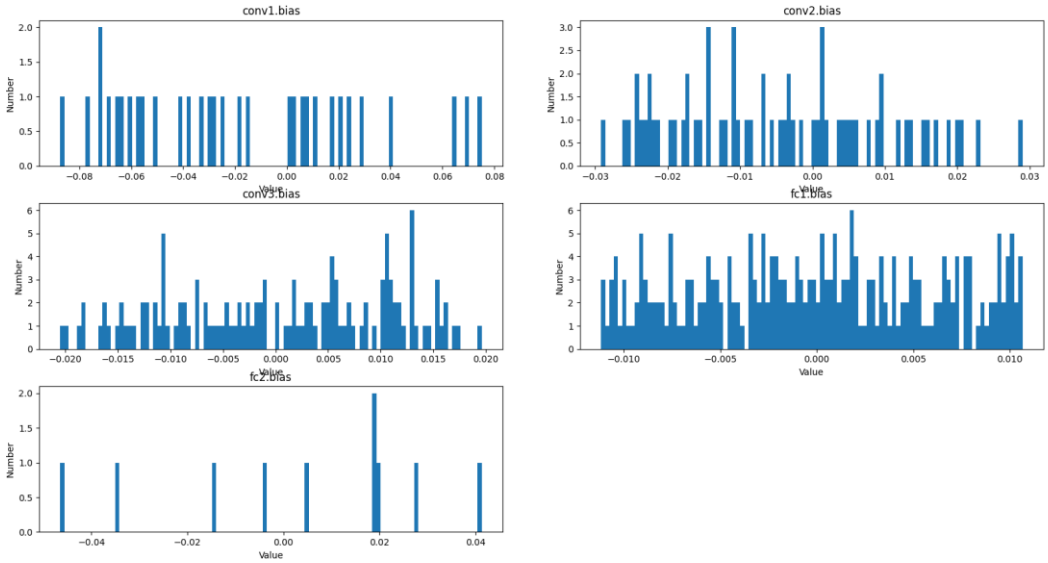
從結果看，*L2 rate* = 5e-2 的情況下，loss 曲線在初期下降得比較快，收斂也早，整體表現穩定。Histogram 顯示權重主要集中在 0 附近，分布窄，代表正則化作用

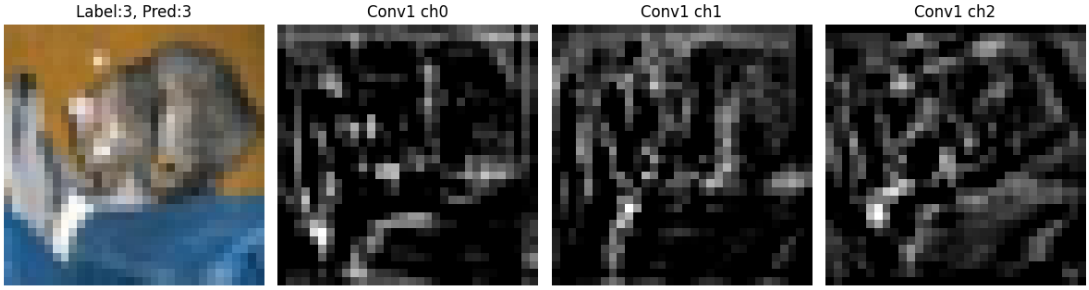
強，模型權重被限制在較小範圍內。

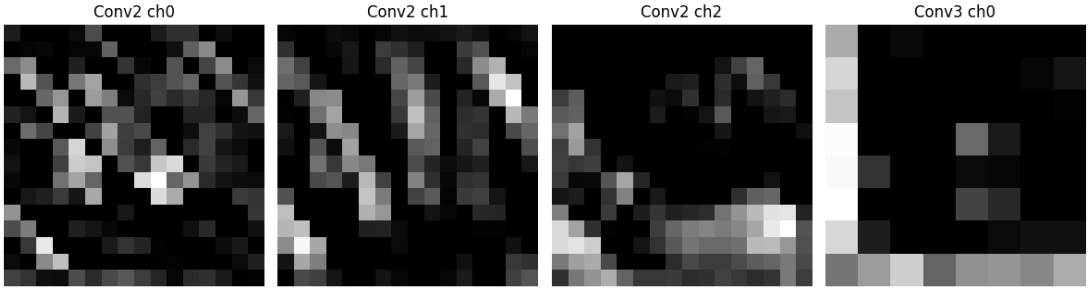
當 *L2 rate* 降到 0 時，loss 曲線波動比較大，收斂速度稍慢，histogram 分布變寬且兩端有明顯尾巴，說明權重值變化更劇烈。這樣的模型雖然初期學習彈性較高，但在測試集容易出現不穩定或過擬合的情況。

II. CIFAR-10

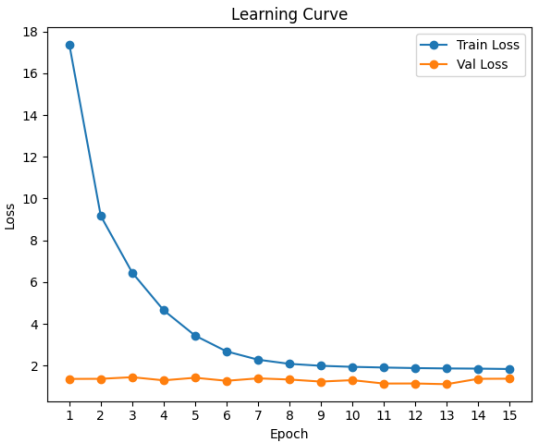


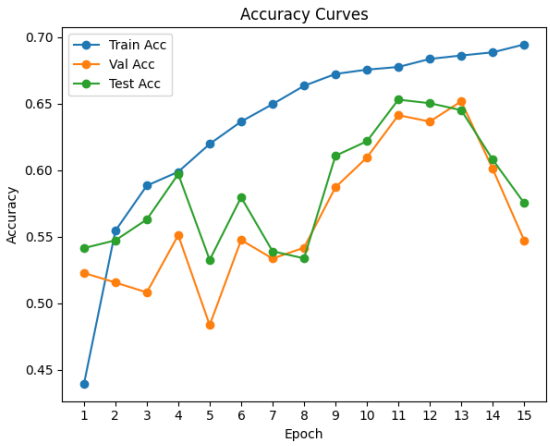


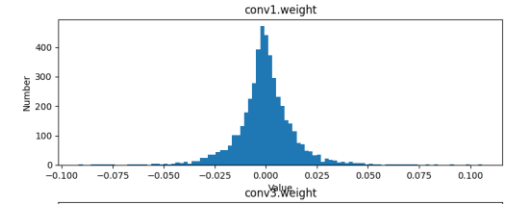


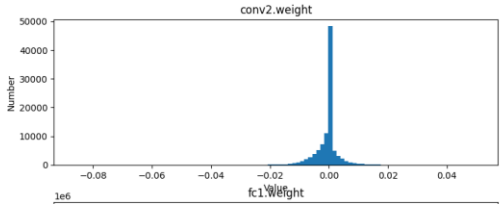


Kernel = 7, *Padding* = 3, *Stride* = 1, *L2 rate* = 5e-2


Learning Curve
Train Loss
Val Loss
Loss
Epoch


Accuracy Curves
Train Acc
Val Acc
Test Acc
Accuracy
Epoch


conv1.weight
Number
Value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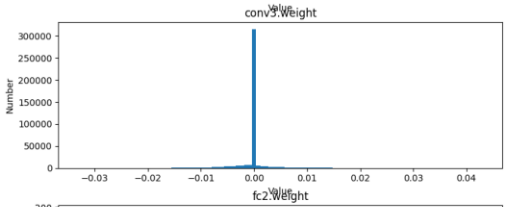
conv3.weight
Number
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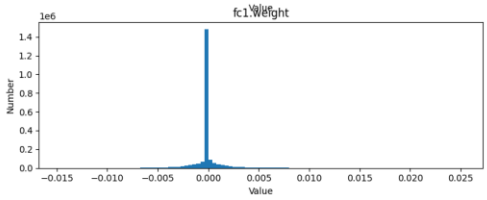
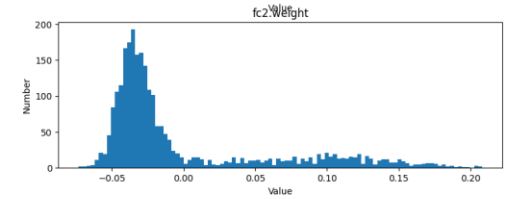

fc1.weight
1e6
Number
Value


fc2.weight
Number
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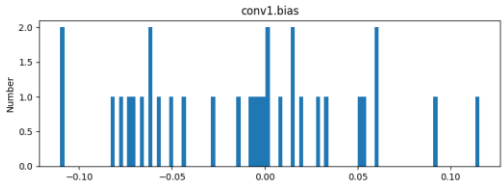
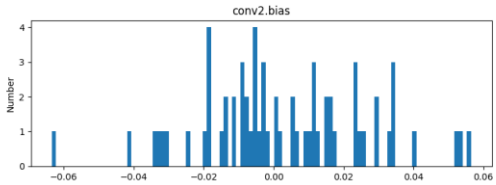
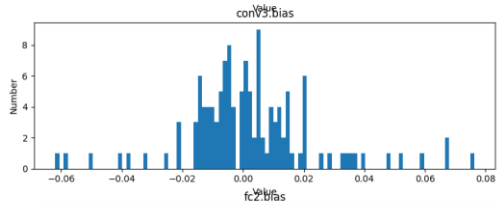

conv1.bias
Number
Value


conv2.bias
Number
Value


conv3.bias
Number
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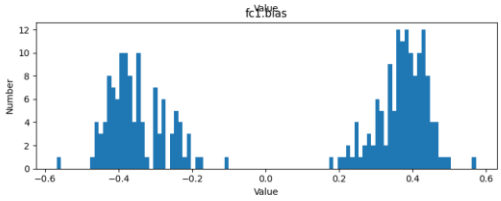
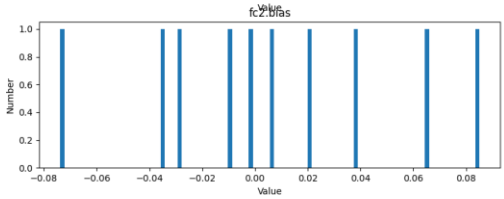
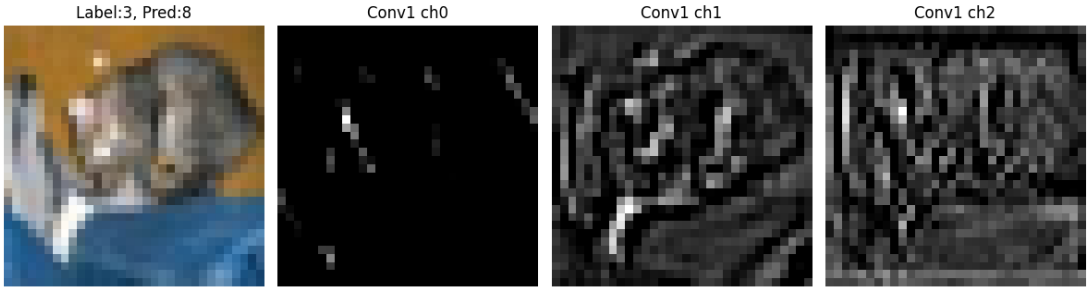
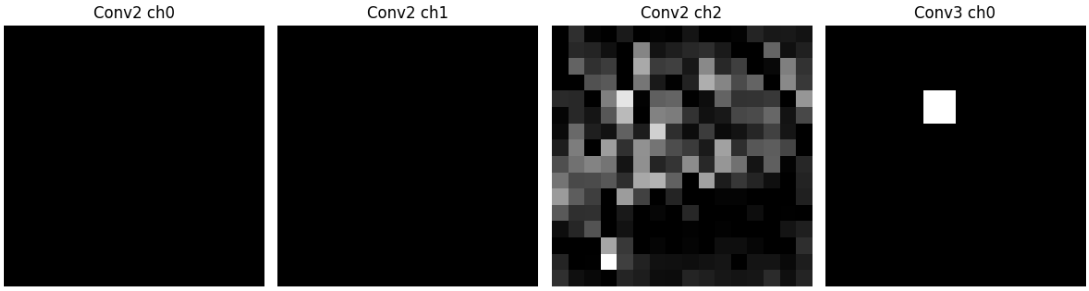

fc1.bias
Number
Value


fc2.bias
Number
Value





Kernel = 7, *Padding* = 3, *Stride* = 1, *L2 rate* = 5e-2

從 histogram 來看，權重分布比 MNIST 寬一些，但仍集中於中間區域，訓練過程中 loss 變化平穩。這樣的 decay 強度讓模型有足夠的權重自由度來捕捉多通道影像的特徵，同時避免權重爆炸。

如果按照 MNIST 的趨勢推估，若 L2 rate 再加大（像 5×e-2），CIFAR-10 的特徵學習可能會被壓得太死，導致模型表現下滑。